Ski Country USA and the Traffic It Creates

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OVERVIEW

In this data mining project, I will be examining historical data of the I-70 corridor from Denver travelling west to Summit County, generally referred to as Ski Town USA. Summit County hosts a high density of high volume ski resorts such as Keystone, Breckenridge, Arapahoe Basin, Copper Mountain, and Vail. Since Denver has a high (and growing) population, and also features an international airport, the I-70 corridor sees a lot of traffic from skiers trying to get to the mountain. This project highlights the key factors when trying to anticipate higher volumes of cars on the road, as well as analyzing historical trends to see how this issue may be growing and during what time of year.

1 Problem Statement and Motivation

Skiing has become an increasingly popular sport in the past decade, and has attracted the attention of corporate entities in the process. The rise in the season pass has allowed more families to make travelling to the mountains on vacation a viable option. This has put additional strain on the resorts and local communities around them, but how has it affected our public infrastructure, specifically the main highway that gets potential skiers to their destination? In this project, I am going to examine the rise in traffic along the I-70 corridor from Denver to Summit County to see what this growth has looked like and what it might develop into in the future. While my main focus will be on the winter months, Colorado is a popular tourist destination year-round, and so I will be looking at all months to assess trends. The ultimate goal is to find some ways around this potential traffic. This could be departing at certain times of day, or certain days of the week. It could also be living in a certain area to avoid the largest slow downs or highest density areas. Where could someone live that wants to balance the proximity of Denver with access to the ski resorts? How might this change in the future if resort access keeps gaining popularity?

The issues examined in this project will be important to anyone that is interested in recreation in the mountains and doesn't already live there. The future trends could be used by local and state governments to assess how we might address the traffic issues and ways to mitigate the amount of cars on the road (alternate routes, public transport, scheduling roadwork in down times/months).

2 Literature Survey

There are a variety of works that are adjacent to this project that aim to inform current drivers. One of those is the popular site CoTrip [1] that informs users of current road conditions, plow activity, and road closures. This is an official site of CDOT and is a very useful resource for real-time data, especially during adverse weather. Another resource that came from CDOT was a report that was put together in 2012: Developing an Active Traffic Management System for I-70 in Colorado [2]. This report goes in depth on road data for the region and tries to map out a process to deal with the rising traffic use, even at that time. This report focuses more on accident data and mitigation techniques to reduce slowdowns on the highway. Things such as variable speed limits based on weather and updated roadwork geometry are big focuses of this report. While this is similar to what I will be looking at, it focuses more on road safety and mitigating crashes, while I hope to see trends in when people are on the road so that you can avoid driving with the bulk of drivers.

Finally, the project *Python Reveals Tunnel Traffic Patterns in Colorado* [3], is the most similar to the project I aim to complete. This project looked at the traffic that passed through the Eisenhower Tunnel at the top of Loveland Pass. This tunnel is the entrance to Summit County, and just below are the towns of Silverthorne and Dillon, where cars will divide depending on which resort they are headed to. My project will focus not just on one spot of I-70, but on the entire stretch of roadway to see which areas see the most traffic and at what times in the day/week/year.

3 Proposed Work

My data for this project will come from the Colorado Department of Transportation and is available on their site [4]. Their historical data tracks sections of road going both directions. The data is sectioned by month and road section. I will need to download just the road sections I am interested in, and then combine the months together to create a holistic data set. For this project, I will be looking at the two continuous counting sites in Clear County, where I-70 travels from Denver to Summit County. The eastern location is in the tunnel in Idaho Springs, and the other is in the Eisenhower Tunnel that goes underneath the continental divide at Loveland Pass. This will be a fairly large dataset, and to ensure that I am able to get the knowledge I want from it, I am going to ensure that the day, month, and year stay attached to the road data that they belong to. My plan is to start from a broad view and narrow down, starting with the yearly trend of traffic data. This will give a good base for how many more people are driving this route over the years. Then I will look at the monthly trends to see what times of year are most popular in order to decide if/how I will divide the data to look at the weekly data from a seasonal standpoint or not. For example, if there are spikes on Thursday afternoon in the summer but not in the winter, I don’t want that affecting any recommendation that I might give for winter travel. Finally, I want to make a decision tree to help users easily see what choices can be made that will lead to more or less traffic. I chose a decision tree because it is easy to interpret even if you are not involved in the probability and data mining sphere. Since the goal of this project is to provide information to a broad group of constituents, a method that is easier to understand is better.

4 Dataset

The data is downloadable from CDOT’s website [4] and features hourly, daily, monthly, and annual summaries. As described earlier, I will start with annual breakdowns to inform decision making and where to investigate further, without being overwhelmed by the amount of data right away. Traffic data for Colorado highways has been tracked on a continuous basis on this site since 1997, including hourly counts, in both directions. This gives me almost 500,000 data points to work with for each section of the highway. The monthly summaries are divided by months as columns, and years as rows. If I am going to examine multiple sections of the highway at once, I will need to join the road sections together based on their time and date. All of the traffic data is an int format as a count of the cars in each direction.

The raw dataset has the following columns: COUNTSTATIONID [int], COUNTDATE [int], COUNTDIR [string], HOUR0-HOUR23[int] (these are individual columns from hour 0 to hour 23, they are all ints and for the sake of length, I will not type them all out), FormattedDate [string]. I am downloading 2 datasets, one from the Eisenhower Tunnel (ET) station, and one from the Idaho Springs (IS) station. The ET dataset contains 19,812 rows, and the IS dataset contains 17,980 rows. Each row currently represents a day and traffic direction and has each hour in the day listed as an individual column.

4.1 Dataset Modification and Preparation

The dataset required some adjustments in order to be analyzed properly and displayed in a helpful way. Any preparation I did in this section happened to both datasets, ET and IS. Firstly, I used pandas [5] to change the COUNTDATE column to be a datetime format. This makes it much easier to parse through the data and extract different years, months and days from this column, instead of having to manually pull that information from the string. Next, I added a column that represented the day of the week. This is an important edition when analyzing this dataset because typically driving traffic changes based on whether it is a weekday or weekend. The column I added was Day of the Week [int] and was a number from 0-6 representing the days of the week. Monday = 0, Tuesday = 1, Wednesday = 2, Thursday = 3, Friday = 4, Saturday = 5, Sunday = 6. This was made much easier since our COUNTDATE column was in a datetime format. Pandas has a function “dt.daysoftheweek” that does this process for us.

Make a new column Daily Totals [int] that adds all of the hour columns together.

My next bit of preprocessing helped me to analyze how many cars came through each station on a monthly and yearly basis. Since the data is broken down by hour, I needed to sum each row to see how many cars came through the count station each day for each direction. I used the pandas .sum() function to sum the totals of all of the hours in each row and create a new column; Daily Totals [int]. I could then use this to look at trends by sorting and summing the Daily Totals row to get my monthly and yearly totals. At this point, the data was ready to be transformed into visualizations that could help me look at trends for monthly, yearly, and day of the week, which we look at later in this report.

After I looked at general trends, I needed to make a few more adjustments to make the dataset ready to make a decision tree. The first thing I did was to stack all of the hourly data into a series. This allowed me to extract the tertiles of the data, as well as the minimum and maximum all time. Using this, I was able to separate the integers in each cell into a string value, either “low”, “medium”, or “high”, to represent how much traffic was on the road at that time. When I make the decision tree, this will allow me to get a clear answer in context for each station. This process is done individually for each count station. This is important because IS receives more traffic than ET, and so we want to make sure our 3 bins represent each dataset equally.

Now that the individual data in the hourly columns has been separated into lower, middle and high thirds, it is time to combine the two datasets into one. This will allow for one decision tree to be made for both datasets at the same time. This is helpful because it simplifies the usage for a consumer, and the count stations can be a factor that we split on, and could have implications for what the traffic will look like. The last bit of preprocessing is to melt the dataset so that each hour is its own row instead of a column. This is needed in order to properly apply the sklearn [4] DecisionTreeClassifier function that will be used to create the tree.

5 Evaluation Methods

When I assess how valuable my findings are, I want to be able to draw insights for each section of I-70. This differs from past research that focuses on real time data or only on a specific point on the road. If I am able to draw different insights for different sections I know my project will have been successful. I will also be able to have different insights depending on what level of macro you want to look at, whether it is a yearly lens, or a monthly lens, my project will attempt to have an insight for each of these scopes and if that is the case, then I know I will have been successful.

I will be building a decision tree in order to quickly and easily interpret the insights gained from the analysis. If a consumer is able to look at the tree and recognize the patterns in the data without too much struggle, then the data mining will have been a success.

6 Tools & Technologies

For my project, I will mainly be using Python and Pandas to manipulate my data, create visualizations, and analyze results. I will also be using the matplotlib library for displaying data. This is especially helpful before any data mining occurs because it will be used to analyze trends over time as the data is processed. Matplotlib is also good because the representations it creates can be easily copied and pasted into other mediums.

The final tool I will talk about is scikit-learn (sklearn) . This is a library built on NumPy, SciPy and matplotlib that focuses on tools for predictive data analysis. This was my first time working with this library and I found it incredibly intuitive and enjoyable to use. Since a decision tree was the final goal, the DecisionTreeClassifier function was critical to the project. However I also found myself using other functions such as LabelEncoder and accuracy_score to help fully round out the project.

I am most comfortable with Python and Pandas and that is where I will start the project and do the bulk of the work.

7 Milestones & Timeline

October 24: Download and connect all relevant yearly summary data. Create a trend line and assess yearly changes.

October 31: Using the yearly data as a guide, download all relevant monthly data. Analyze both trends in months and yearly changes in months to get a scope on what is changing most and where the most interesting changes are occurring. Use these insights to inform the daily breakdowns and make a plan to select certain times of year to assess daily and hourly trends.

November 7: Assess hourly trends of specific times of year influenced by our prior investigations. Begin format of written presentation.

November 14: Upload yearly and monthly date to ArcGIS and format correctly in its relational database.

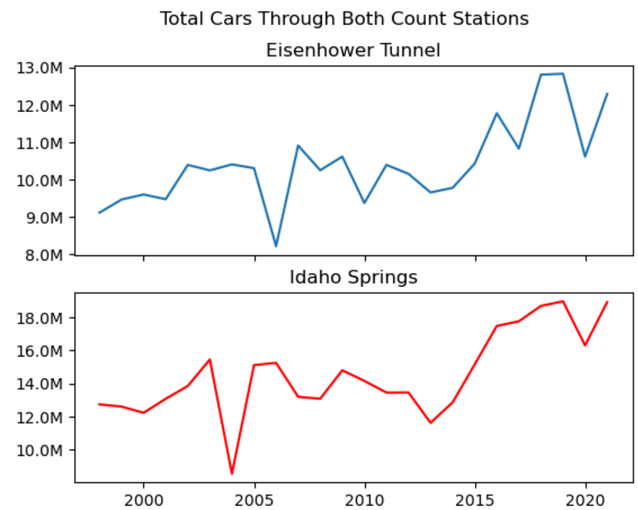
November 21: Upload daily and hourly information into ArcGIS and create visuals that will quickly inform the user how data has changed over time and what insights they can glean from this project.

November 28: Create multiple decision trees that a user can easily interact with and gain insights on best and worst times to drive on I-70 based on where they are going and when.

December 5: Final project presentation

8 Key Results

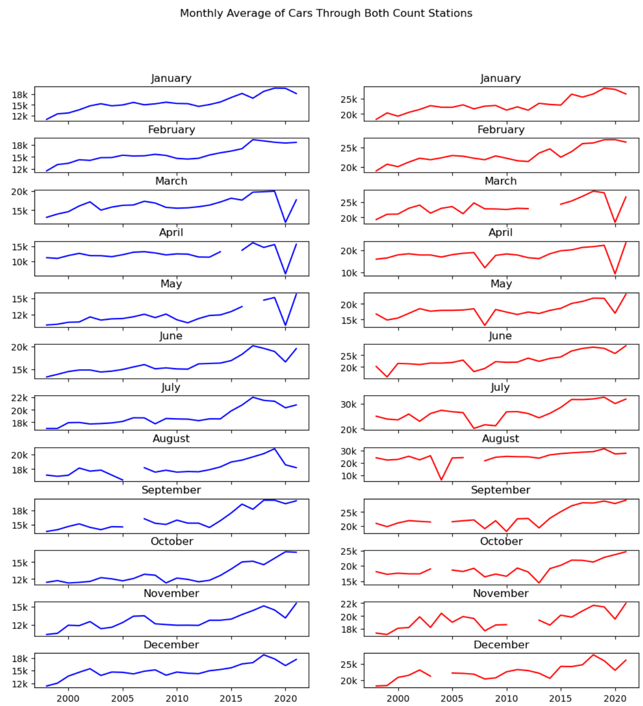
The impetus of this project was that usage of the stretch of I-70 between Denver and Summit County has grown over time. As the data was being processed, it was important to affirm these feelings by backing them up with data. Below is a chart representing the yearly totals of cars through each count station.



There are a couple of interesting things about this data that jump out right away from this chart. Firstly, and most importantly for this project, the trend line for both of the count stations is positive, this confirms the suspicion that the volume of cars along this road has risen over time. We can also see that the IS station is more heavily travelled. This makes sense because this station is closer in proximity to Denver and the large population that resides there.

There are a couple of spikes on both graphs that perhaps we weren't expecting, in 2007 for ET, and 2004 for IS. Perhaps the reasoning for these will come later in the analysis process. There is also a negative spike on both graphs during 2020, we expected this since this was when much of the country was issuing shutdowns and travel was greatly restricted.

Next we should look at our monthly data to see if these trends hold true and if there are other insights to be had.

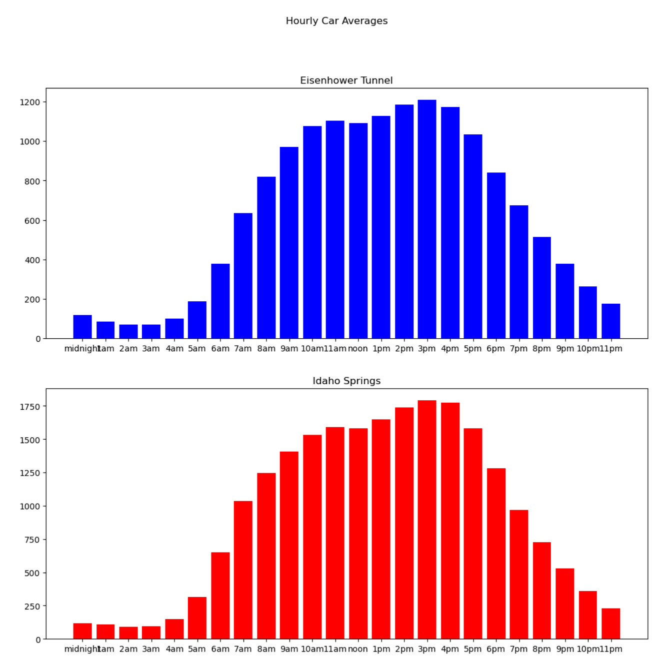


ET is still in blue and is on the left side, while IS is in red and represented on the right side. We get some answers to our earlier questions right off the bat. We can see there are gaps in our data for the ET in 2006 during August and September. This would help explain why there is a negative dip in the yearly graph during that year. The same can be said for the IS graph, in 2004, data for September, October, and December are missing, and there is a steep decline in August. This hints to troubles with the counter during these times and also why the total yearly count could be less than expected in those years.

The trend of increasing volume over time has continued into the monthly breakdown. July also seems to be the most popular month for both count stations. This is interesting because it was not what I was expecting due to the popularity of winter sports in this state. However, July is often when children are out of school and families schedule vacations, which could include driving parts of I-70. July is also often very hot in Denver, and people could be looking to the mountains for recreation in the cooler temperatures that the higher elevations provide. Further analysis would be needed to validate either of those theories.

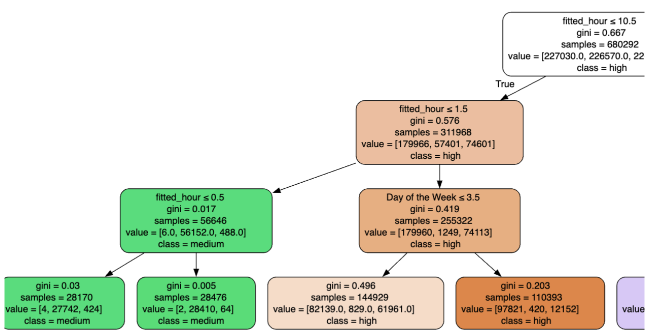
The COVID spike of 2020 is also very apparent in this data. January and February at both stations look as we would expect them to, and then there is a sharp downturn during March due to the travel restrictions and stay at home orders of 2020.

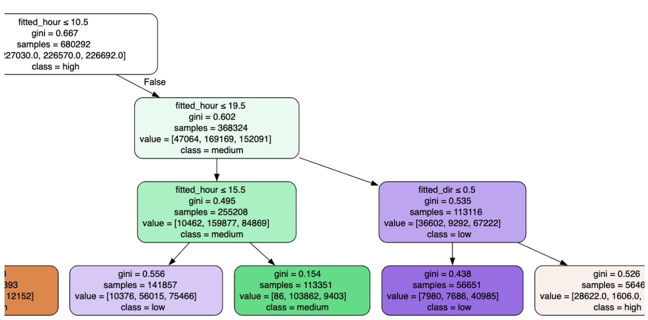
Now that we have looked at long term trends, we can start digging into what is happening on I-70 on a daily basis. For this, we are going to look at the all time hourly averages for each hour in the day for both count stations. We won't be separating by day of the week, but examining all days as one entity.



Looking at the chart above, we can see that there are very few cars on the road during the middle of the night and early morning hours. We can also see that the volume of cars peaks around 3 p.m. for both stations and tapers after that. One interesting thing is that on most roads, there are two peaks, with one centered around the morning rush hour, and another around the afternoon/evening as folks are headed home from work. This spread does not seem to suggest that as much, and instead shows that once traffic picks up on I-70 for the day, it does not slow down until the later evening.

Now that we have looked at an overview of our data and confirmed that it is trending where we thought it was, and it would be useful to see what times are higher or lower volume, it is time to look at our decision tree based around determining what factors go into traffic volume.





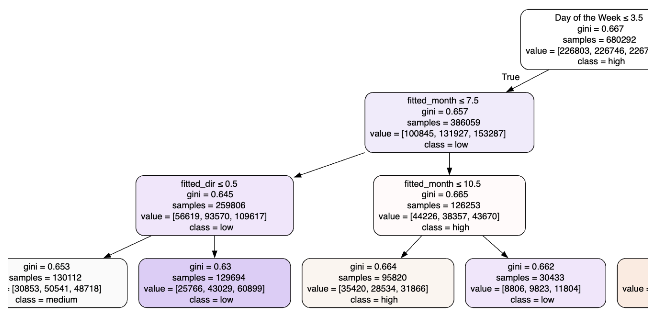
Due to how wide the graph became, it is easier to read if it is split into the two pictures above. This decision tree uses the following features; days of the week, station ID, direction of traffic, hour in the day, and month in the year. The arrows pointing to the left indicate that the statement at the top of the cell is true, while those pointing right indicate false. For our tree, fitted_dir = 0 is eastbound on I-70, and fitted_dir = 1 is westbound. Since we used LabelEncoder to help make our tree, we need to use the following keys for the fitted values for hours and months. For example, if you were driving at 4 p.m. on a Monday in the month of July, your fitted hour to use in the graph would be 8, and fitted month would be 9. From the top node, we would go left because 8 <= 10.5, then we would go right because 8 <= 1.5 is false. Finally, we would go left at the final split because Monday is represented as 0, and 0 <= 3.5. This would tell us that we are likely to see high traffic at this time.

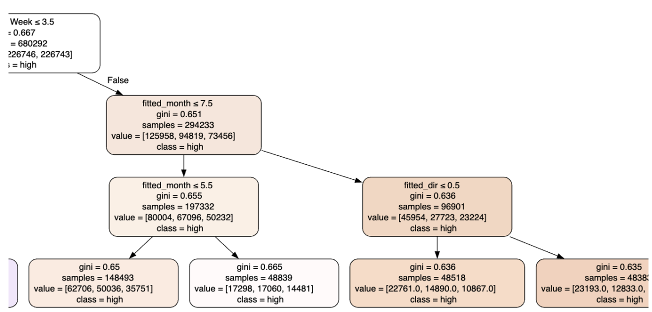
	Hour	fitted_hour
0	HOUR0	0
37794	HOUR1	1
75588	HOUR2	12
113382	HOUR3	17
151176	HOUR4	18
188970	HOUR5	19
226764	HOUR6	20
264558	HOUR7	21
302352	HOUR8	22
340146	HOUR9	23
377940	HOUR10	2
415734	HOUR11	3
453528	HOUR12	4
491322	HOUR13	5
529116	HOUR14	6
566910	HOUR15	7
604704	HOUR16	8
642498	HOUR17	9
680292	HOUR18	10
718086	HOUR19	11
755880	HOUR20	13
793674	HOUR21	14
831468	HOUR22	15
869262	HOUR23	16

	COUNTDATE	fitted_month
0	2001-01-01	0
62	2001-02-01	4
118	2001-03-01	5
180	2001-04-01	6
222	2001-05-01	7
284	2001-06-01	8
344	2001-07-01	9
406	2001-08-01	10
462	2001-09-01	11
501	2001-10-01	1
532	2001-11-01	2
592	2001-12-01	3

An interesting note in this decision tree is that what hour you are driving determines 5/7 choices made in this decision tree. This tells us that it is by far the most important factor when deciding how to avoid traffic. Some interesting things to pull from this regarding traffic times are the low zones. The less surprising take away is that between the hours of 3 and 5 a.m., you are likely to be in “low” traffic volume. The far right of the tree however, tells an interesting story. If you have made it to “fitted_dir <= 0.5”, that means you are driving sometime between the hours of 6 and 9 a.m., and since we splitting on direction of travel, if you are going eastbound (0) you are likely to be in “low” traffic, and if you are headed westbound (1) you are likely to be in heavy traffic.

Since this decision tree was so dominated by the time of day feature, I thought it would be interesting to remove it and build a decision tree just based on day of the week, Station ID, direction of travel, and month of the year to see if there are any other factors that jump out.





At first glance this tree uses more features to distinguish what traffic volume will be, however, it is slightly misleading since ‘gini’ score is high in all nodes. Gini indicates how “pure” a sample set is that the model is making the decision based on. A gini score of 0 means that only one attribute is present when the conditions are satisfied, while a score of 0.667 is a complete variety. The lowest gini on this second tree is

0.63, telling us that the decision it lands on is not as narrow as it could be. Contrasting that with the first tree, whose highest gini on the final tier is 0.555, we can see that the best way to narrow down results is by time of day.

An interesting insight from this second tree is that the entire right side of the tree ends in “high” volume. According to this tree, that means that you are more likely to hit higher volume on the road if you drive on Thursday, Friday, or Saturday, and less likely to hit it on Monday, Tuesday or Wednesday. To get to a “low” traffic volume on the second tree, you would need to be driving on a Monday, Tuesday or Wednesday during September, or during those days headed westbound in any of January, February, March, April, May, October, November, or December. This reaffirms the previous trends we saw in the monthly data when the summer months were a popular time.

9 Applications

The biggest takeaway from this analysis is that the most influence you can exert over how many other people will be on the road at the same time you are, is what time you are driving. Obviously it is not feasible to always commute in the middle of the night, but leaving even an hour earlier could save you much more time if you can avoid the major congestion that comes in the middle of the day. Driving westbound in the mornings is a typical time to encounter traffic, with the peak time of year being in the summer. If you are able, getting into the mountains the night before could save an early rise to “beat the traffic” and make the commute that much more enjoyable.

This stretch of road is continuing to grow in use and popularity, as shown by the yearly and monthly trends. If these continue, we could see the “high volume” time in the middle of the day start to expand as people leave their houses earlier and earlier to beat the traffic, or start commuting in the evening to avoid other drivers. Keeping an eye on how this growth continues and if it ever starts to cap, indicating a reach of maximum volume, will be important for state and local officials. There are plans already in place to try and ease the burden placed on this road and what is key will be how quickly this growth continues.

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